FLO E-Commerce Data Analysis Report

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1. Introduction & Motivation

The retail and e-commerce landscape has seen a massive shift toward digital storefronts over the past decade, making data-driven decision-making essential for anyone seeking to enter or understand the modern marketplace. Our team chose to analyze **FLO Kazakhstan** (flo.com.kz), one of the leading footwear and sportswear retail platforms operating in Central Asia, as the primary subject of our data collection and analysis project.

FLO operates a large-scale online catalog spanning thousands of products across multiple categories, brands, gender segments, and price points. This breadth makes it an ideal candidate for studying how an established retailer structures its inventory, pricing, and promotional strategies.

Our primary motivations for choosing this project are threefold:

Market Research. Understanding how established retailers categorize products, manage large-scale inventories, and structure pricing is foundational knowledge for anyone working in the e-commerce space. By analyzing FLO's catalog systematically, we gain practical insight into how a successful platform operates at scale.

Strategic Planning. Our team has a long-term interest in developing e-commerce solutions for small-scale shops. By scraping and analyzing FLO, we can identify the essential product attributes — such as pricing structures, brand hierarchies, discount patterns, and category organization — that are required to build a competitive retail platform.

Entrepreneurial Insight. This project serves as a technical and analytical foundation for potentially launching an independent online shop in the future. The data we collected and the patterns we discovered will inform decisions about product selection, pricing strategy, and target demographics.

In summary, this project sits at the intersection of technical skill-building and real-world business intelligence. The goal was not merely to collect data, but to extract actionable insights from it.

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2. Data Source Description

Source: FLO Kazakhstan — https://www.flo.com.kz

FLO is a major multinational footwear and sportswear retailer with a strong presence across Kazakhstan and neighboring countries. Its online platform lists thousands of products spanning athletic shoes, casual footwear, children's shoes, and accessories from both international sports brands (Nike, Adidas, Puma, Reebok) and a variety of other labels.

The dataset was constructed by scraping the platform's full product catalog across all categories. Each product record contains the following attributes:

Field	Description
Name	Product name/title
Brand	Manufacturer or brand label
Price	Current sale price in KZT
Old_price	Original price before discount in KZT
Discount	Percentage discount applied
Category	Product category (e.g., running shoes, sneakers)
Model	Specific model identifier
Gender	Target gender (Men, Women, Unisex, Kids)
Color	Product color
Material	Primary external material

The scraping covered pages 2 through 2,000 of the catalog's paginated listing, resulting in a large dataset of real product records available on the platform at the time of collection. This volume provides a statistically meaningful basis for analysis across all major product segments.

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3. Scraping Methodology

The data collection was implemented in Python using the requests and BeautifulSoup libraries, following a two-phase scraping approach that mirrors the structure of the FLO website itself.

Phase 1: Listing Page Scraping

The scraper iterated through paginated listing pages using the URL pattern:

https://www.flo.com.kz/ru/all-categories?page={n}

For each page, the HTML was parsed to extract product cards identified by their CSS class. From each card, the following fields were extracted directly:

- **Name** — from the product__name div
- **Brand** — from the product__brand div

- **Price** — from the product__prices-sale span
- **Old Price** — from the product__prices-actual span
- **Discount** — from the discount badge element
- **Product URL** — the direct link to the individual product detail page

Phase 2: Detail Page Scraping with Multithreading

Several important attributes — specifically **Gender**, **Color**, **Material**, **Model**, and **Category** — are not present on the listing pages. They are only available on individual product detail pages. This required opening each product's URL and parsing its properties section.

To accomplish this while keeping runtime manageable, the team used Python's ThreadPoolExecutor with max_workers=10, enabling up to 10 product detail pages to be fetched concurrently per listing page batch.

responses = list(ThreadPoolExecutor(max_workers=10).map(session.get, product_urls))

For each detail page, the scraper searched for detail-properties__item elements and extracted labeled property pairs (label → value), matching them to the relevant fields by checking Russian-language labels such as Пол: (Gender), Цвет: (Color), and Внешний материал: (Material).

All collected records were appended to a list and exported to a CSV file (flo_products.csv) using pandas with UTF-8-sig encoding to ensure compatibility with Excel and Cyrillic characters.

Session Management

A requests.Session object was used throughout the scraping process. Sessions reuse underlying TCP connections, which reduces overhead when making a large number of sequential and concurrent requests to the same domain.

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4. Data Cleaning Steps with Justifications

After collection, the raw dataset underwent a comprehensive cleaning pipeline to ensure analytical reliability. Each step was applied with a clear rationale.

4.1 Duplicate Removal

Action: Duplicate rows were identified and removed, keeping only the first occurrence.

Justification: Paginated listings on e-commerce platforms sometimes repeat products across pages (e.g., featured or promoted items). Retaining duplicates would skew frequency counts, brand aggregations, and statistical tests.

4.2 Text Normalization

Action: The Name column was stripped of all special characters, retaining only alphanumeric characters and whitespace. The Material column had the abbreviation Комп. (short for "composite") removed for cleaner values.

Justification: Inconsistent formatting in scraped text — such as special symbols, non-breaking spaces, or abbreviations — introduces noise that disrupts grouping, filtering, and display.

4.3 Numeric Type Conversion

Action: The Price, Old_price, and Discount columns were converted from raw strings to numeric types. This involved removing non-breaking spaces (\xa0), the currency label KZT, commas used as decimal separators, and percentage/dash symbols.

Justification: All three columns were scraped as plain text strings. Without conversion, arithmetic operations, statistical summaries, and visualizations would be impossible.

4.4 Missing Value Imputation

Each column with missing values was handled using a strategy appropriate to its data type and business meaning:

Column	Strategy	Justification
Category	Mode (most frequent value)	The dominant category is the most reasonable default for uncategorized products
Model	Backward fill (bfill)	Model names follow a sequential product ordering; adjacent records share similar models
Gender	Filled with "Унисекс" (Unisex)	Products without a specified gender target a general audience
Color	Filled with "Unknown"	No reliable inference is possible; "Unknown" preserves the record without fabricating data
Material	Forward fill (ffill)	Material tends to be consistent within brand/model groups listed sequentially
Old_price	Filled with current Price	If no original price exists, the product likely had no discount and was always at its current price
Discount	Filled with 0	Products with no discount record should not be assumed to have one

4.5 Outlier Handling

Detection: The IQR method was applied to Price, Old_price, and Discount to identify statistical outliers.

Decision: Outliers were **retained** rather than removed. The detected extremes represent real products:

- Very cheap accessories (e.g., 490 KZT)
- High-end premium sneakers (e.g., 99,990 KZT)
- Deep seasonal discounts (up to 88% off)
- Minor markdowns (as low as 5% off)

Removing these rows would eliminate valid product records and distort the true range of FLO's catalog. Instead, **Winsorization (capping)** was applied to limit the influence of extreme values in statistical calculations without discarding the records.

4.6 Feature Engineering

Four new columns were derived from the cleaned data to enrich the analytical potential of the dataset:

- **Savings** — The absolute monetary saving per product (Old_price − Price), giving a concrete figure customers actually save.
- **Brand_Price_Diff** — How far each product's price deviates from its brand's median price, useful for identifying outlier products within a brand.
- **Price_bucket** — Products segmented into four tiers: Budget (0–15,000 KZT), Mid-range (15,000–30,000 KZT), Premium (30,000–50,000 KZT), and Luxury (50,000+ KZT).
- **Is_Premium_Brand** — A binary flag (0 or 1) marking whether the product belongs to Nike, Reebok, or Puma.

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5. Analysis Results with Visualizations

5.1 Descriptive Statistics

The cleaned dataset reveals a wide price range across FLO's catalog. The price distribution is strongly right-skewed (skewness ≈ 1.49), meaning the majority of products fall in lower and mid-range price brackets, with a long tail of premium and luxury items pulling the mean upward. The discount distribution is multimodal, with notable peaks at 0% (no discount), 20%, 40%, and 50%, reflecting FLO's standardized promotional pricing strategy.

5.2 Correlation Analysis

A correlation matrix computed on the numeric columns revealed several important relationships:

- **Price vs. Old_price (r = 0.89):** A strong positive correlation confirming that current sale prices track closely with original retail prices — larger discounts do not fundamentally revalue products.
- **Discount vs. Savings (r = 0.72):** Higher percentage discounts translate directly into larger absolute savings for the customer.

- **Price vs. Discount (r = −0.26):** A meaningful negative relationship — cheaper products tend to carry higher percentage discounts, while expensive items are rarely heavily marked down.
- **Is_Premium_Brand vs. Price (r = 0.53):** Premium brand status is a significant driver of higher pricing.

5.3 Segment-Level Analysis

Average Price by Gender: Men's products average approximately 19,911 KZT, compared to 16,446 KZT for Women's products. Interestingly, Unisex Kids and Unisex categories carry the highest average prices in absolute terms, likely driven by a concentration of premium athletic styles in those segments.

Discounts by Price Segment: Budget items carry the highest average discounts (~24.9%), while Luxury items have the lowest (~6.9%). This reveals a deliberate promotional strategy: FLO uses aggressive discounting to move mass-market volume, while protecting the perceived value of its premium inventory.

Top Brands by Average Price: Nike (~37,250 KZT), Adidas (~33,413 KZT), and Puma (~32,205 KZT) are the most expensive brands on average, driving the upper tail of the price distribution. These brands also account for the premium brand segment analyzed throughout.

5.4 Statistical Hypothesis Testing

Three hypotheses were tested using independent two-sample T-tests (Welch's T-test, unequal variances):

Hypothesis 1: Premium brands are significantly more expensive than non-premium brands.

- Premium average: ~30,600 KZT | Regular average: ~14,133 KZT
- Result: **Accepted** (p-value ≈ 0). The difference is statistically significant. Being classified as a premium brand guarantees a substantially higher price point.

Hypothesis 2: Budget products receive higher percentage discounts than Premium products.

- Budget average discount: ~24.9% | Premium average discount: ~11.8%
- Result: **Accepted** (p-value < 0.05). The store's promotional model is statistically confirmed — high-volume cheap inventory receives aggressive discounts while premium goods do not.

Hypothesis 3: Men's products are priced higher than Women's products.

- Men's average: ~19,911 KZT | Women's average: ~16,446 KZT
- Result: **Accepted** (p-value = 1.61×10^{-27}). The gender-based pricing gap is statistically significant.

5.5 Static Visualizations (Matplotlib & Seaborn)

- Price Distribution — histogram with KDE curve showing the right-skewed nature of pricing
- Discount Distribution — histogram revealing the multimodal promotional structure
- Correlation Heatmap — annotated heatmap of all numeric variable relationships
- Average Price by Gender — bar chart comparing pricing across gender segments
- Average Discount by Price Bucket — bar chart illustrating the inverse discount-price relationship
- Top 10 Brands by Average Price — bar chart highlighting premium brand dominance
- Brand Strategy Bubble Chart — price vs. discount with bubble size representing product count
- Category-Level Bubble Plot — category comparison across average price and old price
- Pair Plot — pairwise relationships across all numeric columns
- Category Share — pie chart of the top 5 most frequent categories

5.6 Interactive Visualizations (Plotly)

- Interactive Scatter Plot — price vs. old price colored by category, with zoom and hover
- Interactive Histogram — price distribution broken down by category
- Interactive Line Chart — trend visualization across sorted price values

6. Key Findings & Insights

The analysis of FLO Kazakhstan's product catalog produced several actionable insights that are relevant both for understanding the platform and for informing future e-commerce strategy:

1. FLO's discount strategy is mass-market focused. The clearest strategic finding is that FLO applies large discounts to budget products and minimal discounts to premium ones. This is not accidental — it reflects a deliberate business model of using promotions to drive turnover in the mid-to-low price segment while protecting brand equity and margin on premium goods.

2. Brand drives price more than any other single factor. The correlation analysis and hypothesis testing together confirm that brand identity is the strongest predictor of price. Nike, Adidas, and Puma products are more than twice as expensive on average as non-premium brands. Any competitive platform must account for this brand-price relationship in its catalog strategy.

3. The price distribution is heavily skewed toward accessible products. Despite the presence of luxury items, the majority of FLO's catalog is concentrated in the Budget and Mid-range tiers. This suggests the platform is primarily positioning itself as an accessible retailer that also carries aspirational premium products.

4. Men's products command a consistent price premium over women's. This gender pricing gap (~3,465 KZT on average) is statistically robust and likely reflects both the product mix (more men's athletic performance shoes) and general market pricing conventions in this category.

5. FLO's promotional calendar is standardized. The multimodal discount distribution — with clear peaks at 0%, 20%, 40%, and 50% — suggests that discounts are applied in predefined tiers rather than on a product-by-product basis. This is useful information for planning competitive pricing on a new platform.

6. Material and color data are underutilized attributes. A significant portion of products had missing color and material fields, requiring imputation. This suggests FLO's internal data hygiene around these attributes has room for improvement — an opportunity for a competitor to differentiate through richer product metadata.

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7. Challenges Faced

Challenge 1: Nested Page Architecture

The most significant technical challenge of this project arose from the structure of the FLO website itself. While basic product information (name, brand, price, discount) was available on the listing pages, several critical attributes — specifically Gender, Color, Material, Model, and Category — were only accessible by navigating into each product's individual detail page.

This required opening a second for loop to follow every product URL scraped from the listing pages, effectively multiplying the number of HTTP requests by the number of products per page. With thousands of pages and dozens of products per page, sequential execution of these detail requests made the scraping process extremely slow — to the point where completing the full dataset would have been impractical within a reasonable timeframe.

Solution: Multithreaded Requests with ThreadPoolExecutor

To resolve this bottleneck, the team implemented Python's concurrent.futures.ThreadPoolExecutor, which allows multiple HTTP requests to be dispatched and processed simultaneously rather than one at a time. With max_workers=10, up to 10 product detail pages were fetched in parallel for each listing batch.

This optimization reduced scraping time by approximately **7 to 8 times** compared to the sequential approach, making full-scale data collection feasible within a realistic project timeline.

Challenge 2: Inconsistent and Missing Data

Scraped web data is rarely clean. FLO's product listings had inconsistent formatting in price fields (non-breaking spaces, mixed decimal separators, appended currency labels), absent attributes on some product pages, and abbreviated material descriptions. Handling these required careful, field-specific cleaning logic rather than a one-size-fits-all approach.

Challenge 3: Outlier Interpretation

A naive approach to outliers would have removed all data points beyond 1.5×IQR. However, the team recognized that extreme values in this dataset represented real products at the edges of FLO's catalog — cheap accessories and expensive premium sneakers — rather than data entry errors. This required a deliberate decision to retain and cap outliers rather than discard them, which involved judgment about the business context rather than mechanical application of a statistical rule.

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8. Bonus:

Bonus 1: Predictive Modeling — Discount Classification

As an extension beyond the core project requirements, a machine learning model was developed to predict whether a product is discounted or not, framing the problem as a binary classification task.

Target Variable: Is_Discounted — a binary flag derived from the Discount column, where any product with a discount greater than 0% is labeled as discounted (1), and the rest as not discounted (0).

Features Used:

- Price — the current sale price of the product
- Brand_Price_Diff — how far the product's price deviates from its brand's median
- Color_encoded — the product's color, label-encoded into a numeric value
- Is_Premium_Brand — binary flag indicating Nike, Reebok, or Puma

Model: A Random Forest Classifier was trained with 100 decision trees on an 80/20 train-test split.

Results: The model achieved an overall accuracy of **70.7%**. It performed better at identifying discounted products (F1-score: 0.76) than non-discounted ones (F1-score: 0.63), which is expected given the class imbalance — discounted products outnumber non-discounted ones in the dataset (932 vs. 633 in the test set).

Feature Importance: The most influential predictor was Brand_Price_Diff (0.423), followed by Price (0.327) and Color_encoded (0.233). Interestingly, Is_Premium_Brand had the least importance (0.017), suggesting that premium brand status alone is a weak predictor of whether a discount is applied — consistent with our earlier finding that premium products are rarely discounted.

Bonus 2: Word Cloud of Product Names

As an additional exploratory step, a word cloud was generated from all product names in the cleaned dataset using the WordCloud library. Each word's size is proportional to its frequency across the entire catalog, providing an immediate visual summary of the most common terms in FLO's product naming conventions.

The visualization was configured with a maximum of 500 words, the viridis colormap, and collocations disabled to ensure each term was counted independently. The resulting cloud reveals two major patterns in how FLO structures its product catalog.

01. Brand Dominance & Market Positioning In-house and exclusive brands outnumber premium labels by volume. The word cloud reveals that Kinetix, Polaris, and Lumberjack are the most frequently appearing brand-related terms, dwarfing internationally recognized names in sheer product count. While Nike, Adidas, Reebok, and Puma are visible, they represent a smaller share of the catalog.

- Top Volume Brands: Kinetix, Polaris, Lumberjack
- Premium Brands Present: Nike, Adidas, Puma, Reebok
- ML Value: Brand identity is a strong categorical feature — our Random Forest model confirmed that Brand_Price_Diff is the single most important predictor (importance: 0.423).

02. Versioned Product Line Strategy FLO relies heavily on iterative model naming conventions. Alphanumeric codes such as 5Fx, 4Fx, 3Pr, and 2Pr appear prominently across the word cloud, indicating that FLO and its suppliers follow a versioned product line strategy. Each iteration likely represents minor design or material updates while retaining brand recognition.

- Common Patterns: XFx series, XPr series, XLi series
- Insight: Product versioning drives catalog volume without requiring entirely new product development.
- ML Value: Model name patterns could be extracted as a feature to capture version-based pricing trends.

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8. Individual Contributions

Yerkebulan Bekbatyr developed the web scraping infrastructure and implemented data cleaning protocols to ensure dataset integrity.

Zhayik Musakhan conducted Exploratory Data Analysis (EDA) to identify statistical trends, distributions, and pricing correlations.

Dinmukhambet Akhmet designed and produced high-fidelity data visualizations to represent market findings and category comparisons.

Yernar Tashkenbay acted as project lead, responsible for technical reporting, final presentation structure, and synthesizing insights.